ICAISC 2027 submission plan — engagement-gated gesture commands on the Go2

Deadline: Sun 15 Nov 2026 (submit by **Fri 13 Nov**, keep the weekend as buffer) Today: Fri 16 Oct → 30 days.

Rules that shape the plan (icaisc.eu)

- Springer **LNAI**, LNCS LaTeX template (`llncs` class on Overleaf). Full papers, **~12 pages**.
- **Double-blind**: anonymised PDF. No author names, no Rysera/WSO2 thanks, **no link to the Vercel site or team GitHub** (use an anonymous repo link, e.g. anonymous.4open.science, or none).
- Camera-ready is LaTeX only, so write in LaTeX from day one.
- Accepted papers must be presented **in person in Zakopane, Poland (20–24 Jun 2027)**. Ask WSO2/Rysera about travel funding early.

What the paper must show

1. Gesture-only control is unsafe in crowds: bystanders trigger it.
2. A per-person fuzzy engagement gate cuts bystander false triggers while keeping commander recall.
3. **IT2 beats Type-1 when perception is uncertain** (robot walking, occlusion, far, turned). This is the claim that makes it an ICAISC paper, so the data must contain enough uncertain cases to test it.

Claim integration novelty only (see CLAUDE.md: Muñoz-Salinas 2005 already did Type-1 fuzzy interest estimation).

Dataset (the critical path)

Item	Target
Participants	**≥ 8 different people**, rotating commander/bystander roles
Locations	2 (Rysera lab + a corridor/lobby)
Takes	9 scenarios (S1–S6, W1–W3) × ~4 reps ≈ **36–40 takes**, 60–90 s each
Episodes	**≥ 150 "not for robot"** and **≥ 150 "for robot"**, about half while walking
Distances	1.5–4 m; some partly occluded and turning
Split	**dev** = dev1 + session-1 pilot takes (tune θ only here); **test** = everything else, never tuned on
Labels	2 annotators label everything independently → report Cohen's κ; disagreements decided by a third
Ethics	signed one-line consent per person; blur faces in figures

Instruction to bystanders: **gesture every ~5 s, hold 2 s, hands visible**. dev1 had only 2–3 bystander gestures, which was the weak point.

Fallback if the dog's WebRTC link fails: mount a phone or webcam on the dog at camera height, record with `--source webcam:N`, and drive with the remote or `--ip --drive`. Same pipeline, same metrics. Say so in the paper.

Evaluation (all offline on logged features, so cheap to rerun)

- **Main table:** false-trigger rate and recall per policy, stationary vs walking, Wilson 95% CIs. **McNemar test** IT2 vs Type-1 on paired episodes.

- **Operating curve:** sweep θ_E and plot false triggers vs recall for all policies. Report false triggers at matched recall (e.g. 90%). Fairer than a single threshold.
- **Where IT2 helps:** split episodes by measured σ (low/high), occlusion and distance.
- **Robustness:** inject noise into logged distance/facing and show IT2 degrades less than Type-1.
- **Ablations:** facing only, distance only, no approach speed; head-only vs body-only facing; FOU scale k.
- **Learned baseline (optional, strong):** logistic regression on the same features, trained on dev only.
- **Closed loop:** unintended robot motions per minute during walking takes, IT2-driven.
- **Runtime:** fps on the laptop (and Jetson if you get one).

Timeline

Dates	Work	Done when
16–19 Oct	Dog connection test; fix go2_link if needed. Consent form. Protocol v2. Book participants and rooms. Overleaf LNCS skeleton.	Live dog camera in `run_live.py`; 8 people booked
20–22 Oct	Pilot session (2–3 takes) = dev split. Freeze all parameters. Write Related Work + Method.	`params.yaml` frozen and committed
23–29 Oct	**Data collection**: 2–3 sessions, ~1.5 h robot time each. Label each session the same day (both annotators).	≥ 300 labelled episodes
30 Oct–4 Nov	Run evaluation, CIs, McNemar, curves, σ-bins, noise test, ablations. Make figures.	All tables/figures generated by one script
5–9 Nov	Write Experiments, Results, Discussion, Intro, Abstract. Full draft by 9 Nov.	12-page draft
10–11 Nov	Internal review by teammates + a lecturer. Fix.	Comments addressed
12–13 Nov	Anonymise, check refs and figures, PDF check, **submit 13 Nov**.	Submission ID received

If semester exams land in this window, move collection earlier, not writing later: the data can't be rushed at the end.

Paper outline (~12 pages LNCS)

1. Introduction (1.5 p): bystander problem, 3 requirements, contributions
2. Related work (1.5 p): fuzzy engagement, addressee estimation, quadruped gesture HRI, IT2 under sensor noise
3. Method (3 p): perception + σ estimation, Layer 1 IT2 FLS (sets, rules, KM), Layer 2 decision, arbitration
4. Experimental setup (1.5 p): robot, participants, scenarios, labelling, metrics, dev/test split
5. Results (2.5 p): main table, operating curve, σ-bins, robustness, ablations, closed loop
6. Discussion and limitations (1 p): monocular distance, one site, small N, single dataset
7. Conclusion (0.5 p)

Ready-made Claude Code prompts

- "Read CLAUDE.md and PLAN.md. Test go2_link against the go2_webrtc_connect examples and fix the API calls."

- "Add a params.yaml for all thresholds; make perception/fuzzy read it; record its hash in every meta.json."
- "Add a second-annotator mode to label_episodes.py (labels_B.json) and compute Cohen's kappa in evaluate.py."
- "Add McNemar (IT2 vs T1), a θ_E sweep with false-trigger-vs-recall curves, σ-bin breakdown, and a noise-injection test to evaluate.py."
- "Add a logistic-regression baseline trained on the dev split only."
- "Convert site/paper_template.html into an LNCS LaTeX paper (llncs), anonymised, with figures from results/."

Risks

Risk	Plan B
WebRTC won't connect	Phone/webcam on the dog (see fallback)
Too few bystander episodes	Extra session; brief bystanders harder
IT2 ≈ Type-1 overall	Report it honestly; focus on high-σ subset and noise test, where the difference should show
Not enough people	Recruit Rysera students/staff; 6 is the minimum
Overfitting thresholds	Tune on dev only, freeze before test sessions, state it